

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

OPERATING & SAFETY MANUAL

TILTING MACHINE

Clutch Plate Positioning & Tilting Station

Manufactured by:

Karthick Engineering & Maintenance Consultants (KEMC)

No 22/50, Manickavasagam Street, Kumaran Nagar, Padi, Chennai – 600050

End User: Valeo India Pvt. Ltd.

Machine Safety Assessment: Independent Safety Consultant

Document Version: 1.0 | Date: February 2026

Safety Assessment Report Date: 16 February 2026

READ THIS MANUAL BEFORE OPERATING THIS MACHINE

Failure to follow safety instructions may result in serious injury or death.

Document Control

Document Title	Operating & Safety Manual – Tilting Machine
Document Number	KEMC-TM-OSM-001
Version	1.0
Date of Issue	February 2026
Prepared by	Technical Documentation Team
Reviewed by	KEMC Engineering

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Approved by	KEMC Management / Valeo Safety Officer
Safety Assessment Reference	Machine Safeguarding Evaluation ADVANCED – 16/02/2026
Next Review Date	February 2027 (or after any machine modification)

Version History

Version	Date	Author	Changes
1.0	February 2026	KEMC / Documentation Team	Initial release

i NOTE: This document is to be kept readily available at the machine workstation at all times. Copies must be provided to all personnel operating, maintaining, or supervising this machine.

Table of Contents

SECTION 1: Machine Identification & Description	5
SECTION 2: Technical Specifications	7
SECTION 3: Standards & Regulatory Compliance	8
SECTION 4: Safety Information & Residual Risks	9
SECTION 5: Guarding & Safety Devices	14
SECTION 6: Machine Layout & Component Identification	17
SECTION 7: Control Panel & HMI Description	19
SECTION 8: Pre-Start Checks	21
SECTION 9: Normal Operating Procedure	23
SECTION 10: Setup & Changeover Procedures	27
SECTION 11: Troubleshooting Guide	30

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

SECTION 12: Cleaning Procedures	33
SECTION 13: Maintenance & LOTO Procedures	35
SECTION 14: Decommissioning	40
ANNEX A: Risk Assessment Traceability Matrix	42
ANNEX B: LOTO Procedure – Pictorial Reference	44
ANNEX C: Inspection & Maintenance Log	45

Image Reference Guide – Manual Insertion Required

The following table lists all 18 images to be inserted manually into this document. Each blue placeholder box in the manual indicates where to paste the corresponding photograph.

Fig	Filename	Description	Insert Location
6.1		Machine Front View – Complete	Section 6 – After component table
7.1		Control Panel (TEKNIC)	Section 7 – Start of section

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Fig	Filename	Description	Insert Location
7.2		HMI Display Screen	<i>Section 7 – Before heading 7.3</i>
5.1		Fixed Mesh Perimeter Guard	<i>Section 5.1 – After heading</i>
5.2		Light Curtain (Vertical)	<i>Section 5.2 – After heading</i>

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Fig	Filename	Description	Insert Location
5.3		E-Stop Button (Red Mushroom)	<i>Section 5.4 – After heading</i>
5.4		Blue Pneumatic Isolation Valve	<i>Section 5.5 – After heading</i>
5.5		Main Power Isolator (Red/Yellow)	<i>Section 5.6 – After heading</i>

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Fig	Filename	Description	Insert Location
A1		FRL Unit (JANATICS) with Gauge	<i>Optional – Section 6 components</i>
A2		DC 24V Solenoid Valve	<i>Optional – Section 6 components</i>
A3		Electric Motor & Worm Gearbox	<i>Optional – Section 6 components</i>
A4		Tilting Fixture with Workpiece	<i>Optional – Section 6 or 9</i>

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Fig	Filename	Description	Insert Location
A5		Pneumatic Clamping Cylinders	<i>Optional – Section 6 components</i>
A6		Earth/Grounding Terminal	<i>Optional – Section 6 components</i>
A7	19_34_22_5.jpeg	Electrical Panel Interior (Safety PLC)	<i>Optional – Section 6 components</i>
A8		Safety Warning Signs (ISO 7010)	<i>Optional – Section 4 or 5</i>

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Fig	Filename	Description	Insert Location
A9		ISO 7010 Pictograms Close-up	<i>Optional – Section 4 or 5</i>
A10		Overall Machine Layout	<i>Optional – Cover or Section 1</i>

i NOTE: HOW TO INSERT: Scroll to the blue placeholder box → Right-click inside the box → Insert > Pictures > Select the image file → Delete the blue placeholder box after image is inserted.

Section 1: Machine Identification & Description

1.1 Machine Identification

Machine Name	Tilting Machine
Machine Type	Assembly Machine – Clutch Plate Positioning & Tilting Station
Manufacturer	Karthick Engineering & Maintenance Consultants (KEMC)

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Manufacturing Date	16 February 2026
Machine Location	No 22/50, Manickavasagam Street, Kumaran Nagar, Padi, Chennai – 600050
End User	Valeo India Pvt. Ltd.
Safety Assessment Report Date	16 February 2026
Safety Consultant	Mr. Muthu Vignesh R – Independent Safety Assessor
KEMC Contact	Mr. Dhinakaran R – Director, kemckemc1@gmail.com, 9884640770
Asset Number	To be assigned by Valeo
Serial Number	To be assigned by KEMC

1.2 Intended Use

The Tilting Machine is designed exclusively for the following purpose:

- Safe loading of assembled clutch plates (Models 230, 280, and 310) onto the tilting fixture using a small overhead crane
- Secure clamping of the clutch plate using pneumatic clamping cylinders
- Controlled tilting of the clamped clutch plate to an ergonomically comfortable angle to enable accurate nut installation using a nut runner
- Returning the machine to the home position and safe unloading of the finished component

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

NOTE: This machine is intended for **INDOOR INDUSTRIAL USE ONLY**. The machine must not be used for any purpose other than described above. Any deviation constitutes misuse and voids all safety evaluations.

1.3 Reasonably Foreseeable Misuse

The following uses are PROHIBITED and constitute misuse of this machine:

- Loading components beyond the specified weight capacity
- Bypassing, defeating, or disabling any safety interlock, guard, or light curtain
- Operating the tilting function while the clutch plate is not fully clamped or is incorrectly positioned
- Manual handling of the clutch plate without the crane
- Performing any maintenance, adjustment, or cleaning while the machine is energised (without LOTO)
- Operating the machine in any mode other than Manual or Automatic
- Allowing untrained or unauthorised personnel to operate the machine
- Using the machine in outdoor, wet, or corrosive environments

1.4 Machine Overview – Process Description

The Tilting Machine processes assembled clutch plates for nut installation in the following sequence:

1. The operator lifts the assembled clutch plate using the overhead crane and places it on the tilting fixture
2. The machine detects the component presence and the operator initiates the cycle via the START button
3. Pneumatic clamping cylinders activate to secure the clutch plate firmly in position
4. The electric motor (via worm gearbox) tilts the fixture to the preset working angle

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

5. The operator installs the nuts using the nut runner tool while the part is held at the tilted angle
6. The operator signals cycle completion; the machine returns to the home/upright position
7. Clamping cylinders release; the operator unloads the finished component using the crane

1.5 Persons Interacting with the Machine

Person Type	Skill Level	Activities
Operator	Instructed	Production, loading/unloading, cycle start, cleaning, changeover
Maintenance Person	Skilled	Troubleshooting, sensor checks, preventive maintenance (every 15 days)
Visitor	Unskilled	Passing by, supervising, observing only – must stay behind safety perimeter

Section 2: Technical Specifications

2.1 Machine Parameters

Electrical Supply	230 V – Single Phase – 50 Hz
Pneumatic Supply	5 Bar (regulated)
Hydraulic System	Not applicable
Operating Modes	Manual / Automatic
Operating Hours per Day	22 hours

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Shifts per Day	3 shifts
Operating Days per Year	230 days
Workpiece	Assembled Clutch Plate
Clutch Plate Models	Model 230, Model 280, Model 310
Cycle Capacity	One clutch plate per cycle
Tilting Mechanism	Electric motor with worm gearbox
Clamping System	Pneumatic cylinders (DC 24V solenoid valves)
Safety Controller	Safety-Rated PLC Controller
Light Curtain Type	Type 4 Safety Rated Light Curtain
Light Curtain Resolution	14 mm
Light Curtain Top Beam Height	1400 mm from ground
Light Curtain Bottom Beam Height	900 mm from ground
Fixed Guard Height	700 mm
Fixed Guard Mesh Gap	20 mm
Control Pendant Brand	TEKNIC
FRL Unit	JANATICS (with pressure gauge, set at 5 bar)

2.2 Workspace & Environmental Conditions

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Installation Environment	Indoor industrial environment
Ambient Temperature	Normal industrial ambient (15°C – 45°C)
Humidity	Normal industrial humidity (non-condensing)
Dust / Oil Mist	Free from excessive dust, oil mist, and corrosive vapors
Floor Space Required	Fixed workstation – defined loading/unloading zone
Clearance Required	Sufficient for tilting movement and crane operation overhead

Section 3: Standards & Regulatory Compliance

This machine has been designed and evaluated in accordance with the following standards and directives:

Standard / Directive	Title & Scope
Machinery Directive 2006/42/EC	European Machinery Directive – overarching legal framework
EN ISO 12100:2010	Safety of Machinery – General Principles for Design – Risk Assessment and Risk Reduction
EN ISO 13849-1:2023	Safety of Machinery – Safety-Related Parts of Control Systems – General Principles for Design
EN ISO 13850:2015	Safety of Machinery – Emergency Stop Function – Principles for Design
EN ISO 13854:2019	Safety of Machinery – Minimum Gaps to Avoid Crushing of Parts of Human Body

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Standard / Directive	Title & Scope
EN ISO 13855:2010	Safety of Machinery – Positioning of Safeguards with Respect to Approach Speeds
EN ISO 13857:2019	Safety of Machinery – Safety Distances to Prevent Hazard Zones Being Reached
EN ISO 14118:2018	Safety of Machinery – Prevention of Unexpected Start-Up
EN ISO 14119:2013	Safety of Machinery – Interlocking Devices Associated with Guards
EN ISO 14120:2015	Safety of Machinery – Guards – General Requirements for Design and Construction
EN ISO 4414:2010	Pneumatic Fluid Power – General Rules and Safety Requirements
EN 60204-1:2018	Safety of Machinery – Electrical Equipment of Machines – Part 1: General Requirements

Section 4: Safety Information & Residual Risks

4.1 Safety Signal Words – Definitions

This manual uses the following signal words to indicate hazard severity:

Signal Word	Meaning
DANGER	Indicates an imminently hazardous situation which, if not avoided, WILL result in death or serious injury.
WARNING	Indicates a potentially hazardous situation which, if not avoided, COULD result in death or serious injury.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Signal Word	Meaning
CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
NOTE	Provides additional information that is important for correct operation but does not involve a hazard.

4.2 Mandatory Personal Protective Equipment (PPE)

All personnel must wear the following PPE at all times when working at or near this machine:

- Safety shoes / Steel-toed footwear (ISO 20345 compliant) – mandatory for all personnel
- Safety gloves – mandatory during loading, unloading, and any manual handling
- Safety glasses – mandatory during all operations
- High-visibility vest – mandatory for visitors and maintenance personnel

⚠ PPE Compliance

Failure to wear the required PPE while operating or maintaining this machine is a serious safety violation. Supervisors are responsible for enforcing PPE compliance at all times.

4.3 General Safety Rules

The following general safety rules **MUST** be observed at all times:

8. Only trained and authorised operators are permitted to operate this machine
9. Read and understand this entire manual before operating the machine

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

10. Inspect the machine before each shift – do not operate if any fault or damage is found
11. Never defeat, bypass, or disable any safety guard, interlock, or light curtain
12. Ensure the light curtain detection zone is completely clear before initiating any cycle
13. Keep all non-essential personnel outside the machine perimeter during operation
14. In case of any emergency, actuate the red E-STOP button immediately
15. Never perform maintenance, cleaning, or adjustment without applying LOTO first
16. Report all near-misses, faults, and injuries to the supervisor immediately
17. Keep the work area clean and free from oil, chips, or any obstruction

4.4 Risk Summary (Per Machine Safety Assessment – 16/02/2026)

The following risks have been identified, evaluated, and risk reduction measures applied per ISO 12100 and ISO 12100 risk assessment methodology:

Risk ID	Hazard Source	Consequence	Task	Risk IN	Risk OUT	Priority
R001	Tilting Motor – Moving Element	Shearing	Production Task	3	<1	3 – Urgent
R002	Pneumatic Clamping Cylinders	Injection Injury	Production Task	3	3*	3 – Urgent*
R003	Electrical Power – Overload	Electrocution	Trouble Shooting	10	<1	1 – Immediate

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Risk ID	Hazard Source	Consequence	Task	Risk IN	Risk OUT	Priority
R004	Tilting Motor – Moving Element	Shearing	Cleaning / Maintenance	3	<1	3 – Urgent
R005	Short Circuit in Control Panel	Electric Shock	Trouble Shooting	10	<1	1 – Immediate
R006	Tilting Motor – Moving Element	Shearing	Visitor	3	<1	5 – Reasonable
R007	Pneumatic Clamping Cylinders	Injection Injury	Gripping Jaw Setup	3	3*	3 – Urgent*
R008	Pneumatic Clamping Cylinders	Injection Injury	Fixture Changeover	3	<1	3 – Urgent

⚠ Open Action – R002 & R007 (Risk OUT = 3)

R002 (Production) and R007 (Gripping Jaw Setup) have a residual Risk OUT of 3 even after all existing measures are applied. The Safety Assessment acknowledges this and requires additional administrative controls: strict adherence to safe work procedures, operator training, and wearing of hand protection (gloves) at all times during these tasks. These risks cannot be fully engineered out with the current machine configuration. Supervisors must ensure compliance.

Section 5: Guarding & Safety Devices

5.1 Fixed Perimeter Guard (SF00)

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

The machine is enclosed by a fixed perimeter mesh guard on all sides except the operator access side. This guard permanently prevents access to the tilting mechanism and pneumatic components during normal operation.

Guard Type	Fixed – Fabricated mesh guard (not interlocked)
Height	700 mm from floor level
Length (per panel)	900 mm approximately
Mesh Gap	20 mm (compliant with ISO 14120 and ISO 13857)
Standard Reference	ISO 14120, ISO 13857
Safety Function	SF00 – Permanently preventing unauthorised entry/access
Status	Implemented and Maintained

DANGER – Do Not Remove Fixed Guard

The fixed perimeter guard must NEVER be removed during operation. Removal of any guard panel must only occur as part of an approved LOTO procedure for maintenance. All guard panels must be reinstated before the machine is returned to service.

Residual Risk – Guard Removal

A residual risk remains if the guard is not replaced after maintenance or repair. Maintenance personnel must verify all guard panels are correctly secured before re-energising the machine.

5.2 Light Curtain – Operator Access Zone (SF13)

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

A Type 4 safety-rated safety light curtain is installed at the operator access opening. This is the PRIMARY active safeguard protecting the operator during the production cycle.

Device Type	Type 4 Safety-Rated Light Curtain
Manufacturer	Third-Party Supplier
Safety Function	SF13 – Safeguarding work areas shared by humans and machines
Required Performance Level	PLr c (per EN ISO 13849-1)
Top Beam Height (from floor)	1400 mm
Bottom Beam Height (from floor)	900 mm
Effective Detection Height	500 mm (900 mm – 1400 mm zone)
Resolution	14 mm
Mounting Orientation	Vertical
Standard Reference	ISO 13855
Status	Implemented and Maintained

The light curtain will immediately halt the machine cycle if any object (including the operator's hand or body) enters the detection zone while the machine is in automatic operation. The machine will not restart until the detection zone is completely clear AND a fresh cycle start command is issued.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

DANGER – Do Not Defeat the Light Curtain

The light curtain must NEVER be defeated, masked, or bypassed under any circumstances. Any attempt to override the light curtain is a safety violation that exposes the operator to serious injury risk. The light curtain must be fully functional before any production cycle is started.

Light Curtain Residual Risks

Two residual risks remain with the light curtain: (1) Inappropriate modification of equipment or fastening solution. (2) Improper location of the detection zone relative to the hazard zone. These must be addressed through periodic inspection and verification of minimum safety distance per ISO 13855.

5.3 Cycle Start Control

The machine uses a single green START-1 pushbutton for cycle initiation. The START button is located on the TEKNIC control pendant at the front of the machine.

Device Type	Single Pushbutton – Cycle Start
Button Colour	Green
Location	TEKNIC control pendant, front of machine
Safety Function	SFXX – Cycle Start Control
Required Performance Level	PLr c
Status	Single button – currently implemented

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

OPEN ACTION – Safety Assessment Reference: Safety Function SFXX

The Machine Safety Assessment (dated 16/02/2026) recommends upgrading the cycle start control to a TWO-HAND ACTUATING CONTROL per ISO 13851. This device would require simultaneous actuation of two separated buttons, ensuring both operator hands are clear of the hazard zone before the cycle begins.

Current Status: Single START button is in use. Two-hand control is PENDING IMPLEMENTATION by KEMC.

Until the two-hand control upgrade is implemented, the operator MUST ensure: (1) Both hands are withdrawn from the hazard zone. (2) The light curtain detection zone is clear. (3) No other personnel are near the machine. ONLY THEN may the START button be pressed.

5.4 Emergency Stop (SF99)

Two emergency stop devices are provided. These are red mushroom-head pushbuttons that must be actuated in any emergency situation to immediately halt all machine motion.

Device 1	Front E-Stop – Red mushroom-head pushbutton on control pendant
Device 2	ON/OFF Switch – Power isolation switch (red/yellow rotary, with lockout provision)
Safety Function	SF99 – Emergency Stop
Required Performance Level	PLr c
Standards	ISO 13850, IEC 60204-1
Status	Both implemented and maintained

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

The E-Stop must be actuated immediately if:

- Any person enters the hazard zone unexpectedly during machine operation
- Any abnormal machine movement, noise, or vibration is detected
- A component drops, shifts, or is incorrectly clamped
- Any guard is removed or damaged during operation
- The operator feels unsafe for any reason

⚠ E-Stop Reset Procedure

After an E-Stop activation, the machine will NOT automatically restart. The cause of the emergency must be identified and resolved first. Then release the E-Stop (twist to unlock), and follow the normal pre-start check procedure before restarting. See Section 8 for pre-start checks.

5.5 Pneumatic Energy Isolation (SF11)

A blue pneumatic isolation ball valve is provided on the air supply inlet of the FRL (Filter-Regulator-Lubricator) unit. This valve must be operated as part of the LOTO procedure to isolate pneumatic energy before any maintenance.

Device	Pneumatic isolation ball valve – Blue handle
Location	FRL panel / air supply inlet
Safety Function	SF11 – Initiating stop and preventing start (pneumatic energy)
Required Performance Level	PLr c

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Standard	ISO 4414
Status	Implemented and Maintained

5.6 Electrical Energy Isolation

A red/yellow rotary main power isolator with lockout provision is located on the machine panel. This must be used as part of the LOTO procedure to isolate electrical energy before any maintenance or troubleshooting.

Device	Red/yellow rotary main power isolator
Lockout Capability	Yes – accepts padlock for LOTO
Safety Function	SF99 – Emergency stop / Energy isolation
Standard	IEC 60204-1
Status	Implemented and Maintained

Section 6: Machine Layout & Component Identification

6.1 Machine Front View – Key Components

The following table identifies all major components visible from the machine front:

Item No.	Component	Description & Location
1	HMI Display	Located on the upper centre of the control panel. Shows process status, sensor alignment guidance, and fault messages.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Item No.	Component	Description & Location
2	Control Panel (TEKNIC)	Main control pendant at front of machine. Houses HMI, mode selector, E-Stop, and START button.
3	Manual / Auto Mode Selector	Rotary key switch on control panel. Selects operating mode (Manual or Automatic).
4	START-1 Button (Green)	Green pushbutton on control panel. Initiates the production cycle. See Open Action in Section 5.3.
5	E-Stop (Red Mushroom Head)	Red mushroom-head pushbutton on front of control panel. Immediately halts all motion.
6	Main Power Isolator (ON/OFF)	Red/yellow rotary switch with lockout provision. Isolates main electrical supply. Lower panel.
7	Light Curtain – Transmitter & Receiver	Type 4 safety light curtain. Vertical pair at operator access opening. Active safety zone: 900 mm – 1400 mm from floor.
8	Fixed Mesh Perimeter Guard	Fabricated steel mesh enclosure. Surrounds machine on all sides except operator access. 700 mm height, 20 mm mesh gap.
9	FRL Unit (JANATICS)	Filter-Regulator-Lubricator panel. Blue pneumatic isolation valve. Pressure gauge (set: 5 bar). Right side panel.
10	Pneumatic Isolation Valve	Blue ball valve handle on FRL unit. Isolates compressed air supply for LOTO.
11	Tilting Fixture / Work Table	Central tilting platform where clutch plate is placed and clamped. Contains pneumatic clamping cylinders.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Item No.	Component	Description & Location
12	Pneumatic Clamping Cylinders	Multiple cylinders inside the fixture. Actuated by DC 24V solenoid valves. Hold the clutch plate during tilting.
13	Electric Motor & Worm Gearbox	Blue motor with worm gearbox. Drives the tilting mechanism. Enclosed inside the mesh guard.
14	Safety-Rated PLC Controller	Yellow safety PLC inside the electrical panel. Processes all safety function inputs and outputs.
15	Earth / Grounding Point	Green EARTH label on panel body. Grounding terminal for electrical safety.
16	ISO 7010 Warning Signs	Yellow/black safety pictograms on machine frame: auto-restart, crush/cut hazard, electrical hazard, LOTO required.

 **IMAGE PLACEHOLDER**

Figure 6.1: Machine Front View – Complete

Insert Image: WhatsApp_Image_2026-02-17_at_19_34_15_12.jpeg

(Machine front view showing control panel, guards, light curtain, FRL unit)

Section 7: Control Panel & HMI Description

 **IMAGE PLACEHOLDER**

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Figure 7.1: Control Panel (TEKNIC)

Insert Image: WhatsApp_Image_2026-02-17_at_19_34_16_13.jpeg
(Control panel showing HMI, mode selector, START button, E-Stop)

7.1 Control Panel Overview

The TEKNIC control pendant is the primary operator interface. All controls required for normal production operation are located on this panel.

i NOTE: Image Placeholder: Insert annotated control panel photograph here, with each control element labelled.

7.2 Control Elements – Description

Control Element	Type / Colour	Function
HMI Display Screen	Digital display	Shows process status, current model selected, sensor alignment guidance during changeover, and fault/alarm messages.
Manual / Auto Selector	Key-operated rotary switch	Selects operating mode. Manual mode for setup, changeover, and maintenance. Automatic mode for production operation.
START-1 Button	Green pushbutton	Initiates the production cycle in Automatic mode. See Open Action – Section 5.3. Do not press until the hazard zone is confirmed clear.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Control Element	Type / Colour	Function
E-Stop Button (Front)	Red mushroom-head pushbutton	Immediately halts ALL machine motion. Actuate in any emergency. Must be released (twist to unlock) before restart.
ON/OFF Power Isolator	Red/yellow rotary switch	Main electrical power isolation. Must be turned OFF and locked for LOTO. Located lower panel / side panel.

IMAGE PLACEHOLDER

Figure 7.2: HMI Display Screen

Insert Image: WhatsApp_Image_2026-02-17_at_19_34_24.jpeg

(HMI display panel showing status information)

7.3 HMI Display Information

The HMI display provides the following information to the operator:

- Current operating mode (Manual / Automatic)
- Current clutch plate model loaded (230 / 280 / 310) – for changeover reference
- Sensor alignment status during gripping jaw setup
- Fault and alarm messages with fault codes
- Cycle status (Idle / Clamping / Tilting / Return / Complete)

i *NOTE: If the HMI shows a fault code or alarm, stop the cycle, note the fault code, and refer to Section 11 (Troubleshooting) before proceeding.*

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Section 8: Pre-Start Checks

Before starting the machine at the beginning of every shift, the operator must carry out the following pre-start inspection. Do NOT start the machine if any item fails the check.

8.1 Pre-Start Inspection Checklist

No.	Inspection Item	Acceptable Condition	Status
1	All fixed mesh guard panels are in position, undamaged, and fully secured	All present & secured	☐ Pass ☐ Fail
2	Light curtain transmitter and receiver are clean, undamaged, and aligned (status indicator green)	Green status indicator	☐ Pass ☐ Fail
3	E-Stop button (front) is released (not depressed) and functioning	Button out / active	☐ Pass ☐ Fail
4	Main power isolator is in the ON position	Rotary switch ON	☐ Pass ☐ Fail
5	Pneumatic supply is connected and pressure gauge reads 5 bar	Gauge at 5 bar $\pm$ 0.2 bar	☐ Pass ☐ Fail
6	Blue pneumatic isolation valve is in the OPEN position	Handle aligned with pipe	☐ Pass ☐ Fail
7	No tools, workpieces, or foreign objects are left inside the machine enclosure	Enclosure clear	☐ Pass ☐ Fail
8	Tilting fixture is in the home (upright) position	Fixture upright	☐ Pass

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

No.	Inspection Item	Acceptable Condition	Status
			☐ Fail
9	Clamping cylinders are in the open/retracted position	Cylinders retracted	☐ Pass ☐ Fail
10	ISO 7010 warning signs are visible, undamaged, and legible	All signs legible	☐ Pass ☐ Fail
11	Work area floor is clean, dry, and free from oil or obstruction	Clean & dry	☐ Pass ☐ Fail
12	HMI display powers up correctly with no fault codes displayed	No fault codes	☐ Pass ☐ Fail
13	Operator is wearing required PPE (safety shoes, gloves, glasses)	Full PPE	☐ Pass ☐ Fail
14	Overhead crane is operational and hook is accessible	Crane serviceable	☐ Pass ☐ Fail

DANGER – Do Not Start If Any Check Fails

If any pre-start check item FAILS, do NOT operate the machine. Inform your supervisor and the maintenance team immediately. Document the failure in the machine log. The machine must not be restarted until the issue is resolved and the failed check passes.

8.2 Machine Energisation Sequence

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

After all pre-start checks are confirmed PASS:

18. Turn the main power isolator to ON position
19. Open the blue pneumatic isolation valve (handle parallel to pipe)
20. Confirm HMI display powers up and shows no fault codes
21. Select the required operating mode (Manual or Automatic) using the mode selector
22. The machine is now ready for operation

Section 9: Normal Operating Procedure

DANGER – Read Before Operating

The tilting motion and pneumatic clamping create shearing and injection hazards. NEVER insert hands or any body part into the machine enclosure during operation. NEVER defeat or bypass the light curtain. Ensure the detection zone is clear before pressing START.

9.1 Step-by-Step Production Cycle (Automatic Mode)

STEP 1 – Mode Selection

Turn the mode selector to AUTOMATIC mode. The HMI will display 'AUTO MODE – READY'.

STEP 2 – Load Clutch Plate onto Fixture

Using the overhead crane and appropriate lifting sling:

23. Attach the lifting sling to the clutch plate as per the approved lifting procedure
24. Slowly lift and manoeuvre the clutch plate over the tilting fixture
25. Lower the clutch plate onto the fixture, aligning it with the locating features
26. Confirm the clutch plate is fully seated and correctly positioned on the fixture

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

27. Remove the lifting sling from the clutch plate
28. Ensure the crane hook is raised clear of the machine enclosure

⚠ Crane Safety

Always use the overhead crane for loading and unloading clutch plates. Never attempt to manually lift or carry the clutch plate into the machine. Ensure the crane operator and machine operator communicate clearly before lifting.

STEP 3 – Confirm Hazard Zone is Clear

Before pressing the START button, confirm ALL of the following:

- Both your hands are completely withdrawn from the machine enclosure
- No tools, slings, or other objects are inside the enclosure
- No other personnel are within the light curtain detection zone (900 mm – 1400 mm from floor)
- The light curtain status indicator is GREEN

STEP 4 – Initiate Cycle (Press START)

With the hazard zone confirmed clear, press the green START-1 button on the control pendant.

! NOTE: OPEN ACTION – Safety Assessment: The Safety Assessment recommends a two-hand cycle start (ISO 13851) to formally ensure both operator hands are clear before the cycle begins. This upgrade is pending. Until implemented, the operator must manually ensure both hands are clear before pressing START (single button).

STEP 5 – Automatic Clamping

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Upon START activation, the safety controller will:

29. Verify the light curtain is unobstructed
30. Activate the pneumatic solenoid valves to extend the clamping cylinders
31. Clamp the clutch plate firmly in position

The HMI will display 'CLAMPING IN PROGRESS' followed by 'CLAMPED – READY TO TILT'.

⚠ Do Not Enter During Clamping

Pneumatic cylinders move rapidly and generate high clamping force. If the light curtain is triggered during clamping, the machine will halt and clamping will stop. The operator must withdraw, resolve the issue, and restart the cycle.

STEP 6 – Automatic Tilting

After successful clamping, the machine automatically tilts the fixture:

32. The electric motor activates via the worm gearbox
33. The fixture tilts the clutch plate to the preset working angle
34. The HMI displays 'TILTING IN PROGRESS' followed by 'AT WORKING ANGLE – READY'

⚠ Tilting Motion Hazard

The tilting motor creates a shearing hazard between the rotating fixture and the fixed machine frame. NEVER reach into the machine enclosure during tilting. If unexpected noise or vibration is detected, press E-STOP immediately.

STEP 7 – Nut Installation

With the clutch plate clamped and tilted to the working angle:

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

35. Pick up the nut runner tool
36. Approach the tilted clutch plate from the front operator access
37. Install the required nuts at all specified positions as per the product work instruction
38. Apply the correct torque as specified in the product drawing or work instruction
39. Visually confirm all nuts are installed and properly seated

The HMI will display the current model and the required torque value for reference.

⚠ Nut Runner Safety

Maintain firm control of the nut runner at all times. Ensure the nut runner cable does not enter the light curtain detection zone or the machine enclosure. Use anti-vibration gloves if the nut runner produces excessive vibration.

STEP 8 – Cycle Completion Signal

After all nuts are installed, signal cycle completion by pressing the START-1 button again (or follow the specific cycle completion input defined by KEMC during commissioning). The HMI will display 'RETURNING TO HOME'.

STEP 9 – Automatic Return & Unclamp

The machine automatically:

40. Tilts the fixture back to the home (upright) position
41. Confirms the home position via limit switch
42. Retracts the pneumatic clamping cylinders to release the clutch plate
43. The HMI displays 'CYCLE COMPLETE – READY TO UNLOAD'

STEP 10 – Unload Finished Component

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Using the overhead crane:

44. Attach the lifting sling to the finished clutch plate
45. Gently lift the clutch plate clear of the fixture
46. Manoeuvre and lower the finished part to the outgoing pallet or station
47. The machine is now ready for the next cycle – return to STEP 2

i *NOTE: The operator should perform a quick visual check of the finished part before placing it on the outgoing pallet. Any part with visibly missing or incorrectly installed nuts must be quarantined and reported to the supervisor.*

Section 10: Setup & Changeover Procedures

DANGER – LOTO Required for Fixture Base Plate Changeover

Fixture base plate changeover (R008) requires full LOTO before any physical adjustment. Refer to Section 13 for the complete LOTO procedure.

10.1 Griping Jaw Setup – Model 280 Adjustment (R007)

When changing production to Model 280, the gripping jaw requires adjustment:

Pneumatic Injection Hazard – R007

Pneumatic clamping cylinders remain energised during gripping jaw setup unless LOTO is applied. The Safety Assessment identifies this as a Risk OUT of 3 (Urgent). Wear gloves at all times during this task. Do not place hands near clamping cylinder actuation paths.

Gripping Jaw Setup Procedure:

48. Switch the mode selector to MANUAL mode

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

49. The HMI will display sensor alignment guidance for the 280 model
50. Using the 3 mm spanner, adjust the bolts and nuts at each stage of the jaw as indicated by the HMI sensor alignment display
51. After each adjustment, check the HMI for 'Sensor Aligned' confirmation
52. Repeat for each jaw stage until all sensors confirm alignment
53. Perform a trial clamp in MANUAL mode to verify correct clamping
54. Switch back to AUTOMATIC mode for production

i NOTE: During gripping jaw setup, follow all LOTO and administrative controls as required. If in doubt about whether the machine is fully de-energised, apply LOTO before touching any clamping component.

10.2 Fixture Base Plate Changeover (R008)

Three base plates are available for Models 230, 280, and 310. Changeover frequency varies from 1 hour to 1 week depending on production demand.

Fixture Base Plate Changeover Procedure:

55. Inform the supervisor that a changeover is being performed
56. Complete all current production cycles and ensure no clutch plate is on the fixture
57. Apply full LOTO as per Section 13 (isolate electrical and pneumatic energy)
58. Remove the locking fasteners from the current base plate
59. Carefully lift and remove the current base plate using appropriate lifting equipment if required
60. Select and position the new base plate for the required model (230, 280, or 310)
61. Secure the new base plate with all fasteners to the specified torque
62. Remove LOTO devices and restore energy sources

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

63. Perform a trial cycle in MANUAL mode to verify correct operation with the new base plate
64. Update the HMI with the new model selection
65. Inform the supervisor that changeover is complete

Section 11: Troubleshooting Guide

⚠ Important

Operators may only perform the operator-level troubleshooting steps listed below. Any troubleshooting that requires opening electrical panels or disassembly of machine components must be performed by authorised maintenance personnel ONLY, with full LOTO applied.

11.1 Troubleshooting Table

Symptom / Fault	Probable Cause	Corrective Action
Machine does not start when START is pressed	Light curtain is obstructed or misaligned	Check light curtain detection zone. Remove any obstruction. Verify green status indicator. Press START again.
Machine does not start when START is pressed	E-Stop is depressed	Check both E-Stop buttons. Release (twist to unlock) if depressed. Resume pre-start check before restarting.
Machine does not start when START is pressed	Mode selector in MANUAL	Switch mode selector to AUTOMATIC mode.
Machine does not start when START is pressed	Pneumatic pressure low	Check FRL pressure gauge – must read 5 bar. Check air supply connection. Report to maintenance if pressure cannot be restored.

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Symptom / Fault	Probable Cause	Corrective Action
Clamping cylinders do not actuate	Solenoid valve fault or low air pressure	Check HMI for fault code. Verify pneumatic pressure at gauge. Stop operation and call maintenance.
Machine tilts but clutch plate is not secure	Clamping not confirmed before tilting	Activate E-Stop immediately. Do not attempt to manually hold the part. Call maintenance to check clamping sensor and cylinder.
Tilting motion is jerky or makes unusual noise	Motor or gearbox fault; foreign object in mechanism	Activate E-Stop. Do not operate. Report to maintenance immediately.
Light curtain status is red / not green	Object in detection zone; dirty optics; misalignment	Check and clear detection zone. Clean light curtain lenses with dry lint-free cloth. Check alignment. If problem persists, call maintenance.
HMI shows fault code	Various – refer to fault code	Note the fault code. Stop the cycle. Refer to HMI fault code list (posted at machine). Call maintenance for fault codes not in the list.
Machine stopped mid-cycle due to E-Stop	E-Stop was actuated	Identify and resolve the cause. Release E-Stop. Perform full pre-start check (Section 8). Restart cycle.
Clamping cylinders remain pressurised after cycle	Solenoid valve stuck; control fault	Do NOT reach into the clamping area. Apply LOTO. Call maintenance.
Unusual vibration during tilting	Clutch plate not correctly seated; motor/gearbox issue	Activate E-Stop. Do not restart. Report to maintenance. Check that clutch

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Symptom / Fault	Probable Cause	Corrective Action
		plate was correctly positioned before next cycle.

DANGER – Electrocutation Risk – R003 & R005

Do NOT open the electrical panel or attempt any electrical troubleshooting unless you are a qualified electrician with appropriate authorisation. Electrical faults carry a Risk IN of 10 (IMMEDIATELY priority). Always apply LOTO before opening any electrical enclosure.

Section 12: Cleaning Procedures

Tilting Motor Hazard During Cleaning – R004

The tilting motor creates a shearing hazard during cleaning near the tilting mechanism. Apply LOTO before cleaning inside the machine enclosure. For external cleaning only (floor around machine, outer panels), LOTO is not required but the machine must be in STOP state.

12.1 Routine Cleaning – Operator (Each Shift)

The following cleaning tasks must be performed by the operator at the end of each shift or as required during the shift:

External Cleaning (Machine in STOP state):

66. Press E-Stop or set mode to MANUAL and ensure machine is stationary
67. Sweep or vacuum chips and debris from the work area floor around the machine
68. Wipe down external surfaces of the machine enclosure with a dry or slightly damp cloth

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

69. Check and clean the light curtain optics with a dry lint-free cloth – avoid abrasive materials
70. Remove any oil spills on the floor immediately using absorbent material
71. Dispose of cleaning waste per factory waste management procedures

Internal Cleaning (LOTO Required):

72. Apply full LOTO as per Section 13 (isolate electrical AND pneumatic energy)
73. Open or remove the required guard panel only after LOTO is confirmed
74. Remove chips, oil, and debris from the fixture, clamping cylinders, and tilting mechanism using brush or vacuum
75. Wipe down fixture surfaces with a clean cloth
76. Check clamping cylinder rods for cleanliness and damage – clean with dry cloth
77. Do NOT use high-pressure air to blow chips as this can project debris
78. Replace all guard panels and confirm secure
79. Remove LOTO and restore energy before resuming production

12.2 Lubrication (Maintenance – Every 15 Days)

Lubrication is performed by the authorised maintenance person as part of the preventive maintenance programme (PM every 15 days):

- Lubricate worm gearbox per manufacturer specification
- Apply grease to tilting mechanism pivot points as required
- Check and lubricate FRL lubricator level – top up if required with the recommended pneumatic lubricant
- Check cylinder rod surfaces – apply a light film of compatible lubricant if required

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

i NOTE: All lubricants must be compatible with pneumatic seals and food-grade requirements if applicable. Use only approved lubricants as specified by KEMC. Record all lubrication in the maintenance log (Annex C).

Section 13: Maintenance & LOTO Procedures

DANGER – LOTO is Mandatory

Lockout / Tagout (LOTO) must be applied before ANY maintenance, repair, cleaning inside the enclosure, or adjustment on this machine. Failure to apply LOTO when required may result in DEATH or serious injury.

13.1 Energy Sources on This Machine

Energy Source 1 – Electrical	230 V Single Phase AC via main power isolator (red/yellow rotary switch)
Energy Source 2 – Pneumatic	5 bar compressed air via blue pneumatic isolation ball valve on FRL unit
Stored Energy – Pneumatic	Pressure trapped in cylinders and pipework after isolation – must be vented
Stored Energy – Gravity (Tilting)	If fixture is at tilted position, gravity load remains. Machine must be in home (upright) position before LOTO.

13.2 LOTO Procedure – Step by Step

Step	Action Required	Verification
1	NOTIFY: Inform all personnel in the area that the machine is being isolated for maintenance. Ensure no one is inside the machine enclosure.	<i>Verbal notification complete</i>

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Step	Action Required	Verification
2	STOP: Press E-Stop or set mode selector to MANUAL. Ensure all machine motion has completely stopped.	<i>Machine fully stopped</i>
3	RETURN TO HOME: If the fixture is tilted, use MANUAL mode to return it to the upright home position before proceeding.	<i>Fixture at home position</i>
4	ISOLATE ELECTRICAL: Turn the main power isolator (red/yellow rotary) to the OFF position.	<i>Isolator in OFF position</i>
5	LOCKOUT ELECTRICAL: Apply your personal padlock to the main power isolator lockout hasp. Retain the key.	<i>Padlock applied – key retained</i>
6	TAGOUT ELECTRICAL: Attach a 'DO NOT OPERATE – MAINTENANCE IN PROGRESS' tag to the isolator.	<i>Tag attached and legible</i>
7	ISOLATE PNEUMATIC: Turn the blue pneumatic isolation ball valve to the CLOSED position (handle perpendicular to pipe).	<i>Blue valve closed</i>
8	VENT STORED PRESSURE: Manually vent any trapped pressure from cylinders per the cylinder venting procedure. Confirm pressure gauge reads zero.	<i>Gauge reads zero / 0 bar</i>
9	VERIFY ZERO ENERGY: Attempt to start the machine by pressing START. Confirm machine does NOT start.	<i>Machine does not start on START press</i>
10	COMMENCE MAINTENANCE: Only after all steps above are confirmed – proceed with maintenance.	<i>Authorised maintenance begins</i>

13.3 Restoring Energy After Maintenance (LOTO Release)

80. Confirm maintenance is fully complete and all tools/parts are accounted for

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

81. Ensure all guard panels are reinstated and fully secured
82. Notify all personnel that LOTO is being released and the machine is about to be re-energised
83. Confirm no one is inside the machine enclosure
84. Open the blue pneumatic isolation valve (handle parallel to pipe)
85. Remove your personal padlock and LOTO tag from the main power isolator
86. Turn the main power isolator to ON
87. Perform the full pre-start check (Section 8) before resuming production

DANGER – Multiple LOTO Devices

If more than one person is working on the machine, EACH person must apply their own personal padlock and tag. Only the person who applied their padlock may remove it. The machine must not be re-energised until ALL padlocks are removed.

13.4 Preventive Maintenance Schedule

Maintenance Task	Frequency	Responsible	Remarks
Pre-start inspection (Section 8)	Every shift	Operator	Record in shift log
Light curtain optics cleaning	Every shift	Operator	Use dry lint-free cloth only
Floor and external cleaning	Every shift	Operator	Remove chips and oil spills
Internal fixture cleaning	Weekly or as needed	Operator / Maintenance	LOTO required

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Maintenance Task	Frequency	Responsible	Remarks
Lubrication – gearbox, pivots, FRL	Every 15 days	Maintenance	<i>Use approved lubricants. Record in Annex C</i>
Sensor and limit switch check	Every 15 days	Maintenance	<i>Check alignment and response. LOTO required</i>
Cylinder condition check	Every 15 days	Maintenance	<i>Check for leaks, rod damage, seal wear</i>
Light curtain alignment & safety distance check	Monthly	Maintenance / Safety	<i>Verify per ISO 13855</i>
E-Stop function test	Monthly	Maintenance	<i>Actuate and verify machine halts</i>
SRPCS (Safety PLC) diagnostic check	Monthly	Maintenance	<i>Check status LED and fault log</i>
Full electrical inspection	Annually	Qualified Electrician	<i>IEC 60204-1 compliance</i>
Full machine safety re-evaluation	Every 3 years or after modification	Safety Officer / Independent Assessor	<i>Update RA if machine is modified</i>

Section 14: Decommissioning

14.1 Temporary Shutdown

For temporary shutdown at the end of a shift or for maintenance:

88. Complete the current cycle and ensure no clutch plate is on the fixture

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

89. Set the mode selector to MANUAL
90. Confirm the tilting fixture is in the home (upright) position
91. Press E-Stop to halt the machine
92. Turn the main power isolator to OFF
93. Close the blue pneumatic isolation valve
94. Record the machine shutdown in the shift log

14.2 Planned Decommissioning

When the machine is to be decommissioned permanently or relocated:

95. Inform KEMC and Valeo management and obtain written authorisation
96. Complete all current production and clear the machine
97. Apply full LOTO as per Section 13
98. Disconnect all energy sources – electrical and pneumatic
99. Drain and vent all pneumatic circuits
100. Remove and retain all safety devices (light curtain, SRPCS) for reuse or proper disposal
101. Dispose of all mechanical, electrical, and pneumatic components per applicable environmental and waste management regulations
102. Retain all documentation (this manual, Safety Assessment report, maintenance logs) for a minimum of 10 years

i *NOTE: Any modification or recommissioning of this machine after decommissioning requires a new risk assessment to be performed by an independent qualified body before the machine is returned to service.*

Annex A: Risk Assessment Traceability Matrix

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

This matrix traces each identified risk from the Machine Safety Assessment (16/02/2026) to the corresponding sections in this manual where the risk is described and the risk reduction measures are documented.

Risk ID	Hazard & Task	Consequence	Risk IN	Risk OUT	Manual Section (Risk Reduction)
R001	Tilting Motor – Production Task	Shearing	3	<1	Sec 5.2 (Light Curtain), Sec 5.3 (Cycle Start), Sec 5.4 (E-Stop), Sec 5.1 (Fixed Guard), Sec 9
R002	Pneumatic Cylinders – Production Task	Injection	3	3*	Sec 5.2 (Light Curtain), Sec 5.3 (Cycle Start), Sec 5.5 (Air Isolation), Sec 9 – OPEN RISK
R003	Electrical Power – Troubleshooting	Electrocution	10	<1	Sec 5.6 (Electrical Isolation), Sec 13 (LOTO), Sec 11
R004	Tilting Motor – Cleaning / Maintenance	Shearing	3	<1	Sec 5.2 (Light Curtain), Sec 5.4 (E-Stop), Sec 5.5 (Air Isolation), Sec 12, Sec 13 (LOTO)
R005	Short Circuit – Control Panel – Troubleshooting	Electric Shock	10	<1	Sec 5.6 (Power Isolation), Sec 13 (LOTO), Sec 11
R006	Tilting Motor – Visitor	Shearing	3	<1	Sec 5.1 (Fixed Guard), Sec 5.2 (Light Curtain), Sec 4.3 (Safety Signs)

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Risk ID	Hazard & Task	Consequence	Risk IN	Risk OUT	Manual Section (Risk Reduction)
R007	Pneumatic Cylinders – Gripping Jaw Setup	Injection	3	3*	Sec 5.2, 5.3, 5.4, 5.5, Sec 10.1 – OPEN RISK
R008	Pneumatic Cylinders – Fixture Changeover	Injection	3	<1	Sec 5.5, 5.6, Sec 10.2, Sec 13 (LOTO)

⚠ Open Actions – R002 & R007

* Risk OUT = 3 for R002 and R007 indicates these risks remain at 'URGENT' level even with all measures applied. The two-hand cycle start (SFXX) is recommended by the Safety Assessment but pending implementation. Administrative controls (training, SWP, PPE) are essential until the engineering upgrade is complete.

Annex B: LOTO Procedure – Pictorial Reference

The following pictorial summary must be posted at the machine workstation. It provides a quick visual reference for the LOTO sequence.

i NOTE: Image Placeholder: Insert a custom LOTO pictorial card here showing: (1) Main power isolator – OFF + padlock. (2) Blue pneumatic valve – CLOSED. (3) Vent residual air – pressure gauge at zero. (4) Verify zero energy – press START, machine must not respond. (5) LOTO tag with operator name, date, and signature.

B.1 LOTO Equipment Required

Padlock (1 per person)	Personal padlock with unique key – must be retained by the person performing maintenance
-------------------------------	--

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

LOTO Tag	'DO NOT OPERATE – MAINTENANCE IN PROGRESS' tag – name, date, and reason to be filled
LOTO Hasp	Multi-lock hasp on main power isolator – allows multiple padlocks for multiple workers
Energy Isolation Points	1. Main power isolator (electrical) – 2. Blue pneumatic ball valve (pneumatic)

B.2 LOTO Tag Information

Each LOTO tag must show:

- Name of person performing maintenance
- Date and time LOTO was applied
- Reason for isolation
- Expected completion time
- Supervisor notified – name

Annex C: Inspection & Maintenance Log

This log must be maintained at the machine. All inspections, maintenance activities, faults, and repairs must be recorded here.

Date	Time	Activity / Observation	Action Taken	Technician Name	Sign

NO. 22/50, MANICKAVASAGAM STREET, KUMARAN NAGAR, PADI, CHENNAI – 600050
TAMIL NADU, INDIA

KARTHICK ENGINEERING & MAINTENANCE CONSULTANTS

Date	Time	Activity / Observation	Action Taken	Technician Name	Sign